PETROLEUM AND LAND-USE FACTOR

Gas Stations

The visible pumps are only part of the site; underground tanks, lines, vents, spills, historic operations, and groundwater plumes can outlast the business.

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1 Understand the factor

What it is - and why it matters

Gas stations store gasoline and sometimes diesel in underground storage tanks connected to fill pipes, product lines, dispensers, vapor-recovery equipment, and vents. Routine operations create vehicle traffic and fuel vapors; releases can occur from tanks, piping, overfills, spills, corrosion, or old systems. Petroleum contains chemicals such as benzene, toluene, ethylbenzene, xylenes, and fuel additives. A closed station may still have residual contamination or monitoring wells even after tanks are removed.

How risk can reach a household

- Leaked fuel can move through soil and groundwater, sometimes beyond the parcel and beneath nearby buildings.
- Volatile compounds can migrate as soil gas and enter structures through cracks, utility penetrations, sumps, or crawlspaces.
- Vehicle refueling and tank deliveries create short-term vapor, idling, traffic, lighting, noise, and collision risks.
- Fire, tanker spill, or dispenser impact can create an acute emergency and require evacuation or road closure.

What people often miss

- A 'case closed' cleanup may leave residual contamination under risk-based controls, and a closure date does not prove unrestricted use.
- Groundwater flow and utility trenches can carry impacts in directions that do not match straight-line distance.
- The current station may have modern double-wall tanks while a former station nearby had older single-wall systems.
- Fuel odor indoors warrants prompt investigation, but odor can also come from vehicles, stored products, or plumbing and does not identify concentration.

? Five checks before buying

1. Search EPA UST Finder and state cleanup databases for active and former stations, tank status, releases, plume maps, monitoring, and closure letters.
2. Review historical land use for old filling stations, auto repair, dry cleaning, and bulk fuel storage at the property and adjoining parcels.
3. For a home near a plume, have an environmental professional evaluate groundwater depth, soil gas, foundation type, and vapor-intrusion data.
4. Visit during deliveries and peak traffic to assess odors, idling, lighting, access conflicts, and noise.
5. Ask the insurer about pollution, fire, explosion, vehicle impact, remediation, loss of use, and known environmental conditions.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 18

H Health impacts

- High acute vapor exposure can cause headache, dizziness, nausea, and irritation; fire or explosion can cause burns and smoke injury.
- Traffic-related pollution near busy stations may overlap with roadway exposure and can be more important than the station alone.
- Benzene is a known human carcinogen, but health risk depends on measured concentration and duration through a complete pathway.
- Odor, overnight activity, and safety concerns can affect sleep and stress even where indoor concentrations are below health benchmarks.

P Property impacts

- A petroleum plume can require monitoring wells, vapor barriers, sub-slab depressurization, soil removal, or restrictions on construction and groundwater use.
- Light, traffic, litter, noise, crime perception, and commercial expansion can affect neighborhood use and resale preference.
- Cleanup access and equipment can disrupt yards, parking, roads, and utilities for years.
- A tanker or dispenser incident can damage structures and vehicles and deposit fuel or firefighting residue.

I Insurance implications

- Standard homeowners insurance commonly excludes pollution and pre-existing or gradual petroleum contamination.
- State tank-cleanup funds and responsible-party insurance may finance remediation, but they do not guarantee compensation for every homeowner impact.
- Sudden fire or vehicle damage may be covered while soil, groundwater, vapor testing, and diminished value remain limited.
- Known vapor systems or land-use restrictions should be disclosed and discussed with the carrier and lender.

Possible long-term health implications of buying nearby

- Long-term health concern is greatest where laboratory data confirm vapor, drinking-water, or soil exposure, not merely where a station is visible.
- Children and residents with respiratory disease may be more affected by combined traffic, odor, and indoor-air conditions.
- A modern compliant station without a release has a different risk profile from a former site over a migrating plume.
- Vapor-mitigation systems are effective only when fans, alarms, seals, and discharge points remain functional.

Evidence lens

Petroleum leakage and vapor intrusion are established mechanisms. Site-specific health conclusions require current environmental data, building-pathway evaluation, and comparison with regulatory screening levels.

Plan more carefully for

Children · Older adults · Pregnancy · Heart/lung conditions · Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

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V Verify the actual risk

- Search active and historical UST records, cleanup cases, plume maps, monitoring wells, and closure conditions for surrounding parcels.
- Obtain professional vapor review when the home is over or near a petroleum plume or has persistent fuel odor.

P Protect the property

- Maintain any vapor barrier or depressurization system, keep foundation penetrations sealed, and protect monitoring wells.
- Do not disturb contaminated soil or install a groundwater well without agency approval and a soil-management plan.

M Monitor and respond

- Report persistent indoor fuel odor, visible spills, sheen, dead vegetation, or vapor-system alarms promptly.
- During a spill or fire, avoid ignition sources and follow official evacuation, shelter, and cleanup instructions.

I Prepare insurance records

- Clarify pollution, fire, explosion, vehicle, testing, remediation-system, and loss-of-use coverage.
- Keep agency closure documents, sampling reports, mitigation maintenance, photographs, and incident records.

REAL-LIFE EXAMPLE

Santa Monica MTBE Drinking-Water Contamination

Detected in the 1990s

The City of Santa Monica shut drinking-water wells after methyl tert-butyl ether, a gasoline additive known as MTBE, contaminated groundwater. EPA actions and litigation addressed leaking underground fuel systems and restoration of the city's water supply.

Why it matters: The case showed that numerous petroleum releases can affect municipal groundwater, close wells, and create long-term treatment and liability costs beyond the footprint of any single station.

Open official incident record - epa.gov >

Questions to resolve before making a decision

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2. Review historical land use for old filling stations, auto repair, dry cleaning, and bulk fuel storage at the property and adjoining parcels.
3. For a home near a plume, have an environmental professional evaluate groundwater depth, soil gas, foundation type, and vapor-intrusion data.
4. Visit during deliveries and peak traffic to assess odors, idling, lighting, access conflicts, and noise.

Official sources and mapping tools

EPA underground storage tanks - epa.gov >

EPA UST Finder - epa.gov >

EPA vapor intrusion - epa.gov >